

RHEMA ERP FINANCE MODULE

REVISED REALISTIC 4.5-MONTH MVP TIMELINE

ASP.NET Core Web API - Properly Scoped for Production Readiness

EXECUTIVE SUMMARY

This revised timeline reflects a realistic 4.5-month development plan based on comprehensive analysis of the complete Finance Module workflows. The scope has been carefully refined to deliver a production-ready MVP that can genuinely close books monthly and serve external customers, while deferring complex features that aren't critical for initial launch.

Key Changes from Original Timeline:

- **Added Cash Management/Treasury:** 15 days for bank reconciliation, cash forecasting, imprest accounts, and cheque management
- **Proper Fixed Assets Scoping:** Reduced to single-book, core depreciation methods; deferred componentization, CWIP automation, and IFRS 16
- **Added Basic Tax Management:** VAT/GST handling and withholding tax calculations (8 days)
- **Enhanced Currency Revaluation:** Includes IAS 21 compliance with automatic reversal mechanism (10 days)
- **Added Period Close Procedures:** Essential for closing books monthly (6 days)
- **Realistic Effort Estimates:** Based on actual workflow complexity documented in my specifications

MVP Success Criteria (Unchanged)

- Close books monthly for internal operations with full audit trail
- Market-ready REST API for client use with comprehensive documentation
- Complete financial cycle: GL, AP, AR with multi-currency support
- Cash management including bank reconciliation and cash flow forecasting
- Basic inventory management with WAC costing
- Budget allocation and tracking with variance analysis
- Core fixed assets with straight-line and reducing balance depreciation

REFINED MVP SCOPE

IN SCOPE - MVP v1.0

Chart of Accounts & General Ledger

- Segmented accounts (up to 20 segments) with lookup validation
- Reporting dimension flags (controls which segments appear in reports)
- Multi-currency GL accounts with parallel currency tracking
- Currency link protection (prevents removal with transaction history)
- IFRS/Base/Local classification for multiple reporting frameworks
- Journal entries with maker-checker via external approval API
- Trial Balance, Income Statement, Balance Sheet with drill-down

Multi-Currency Management

- Base currency and foreign currency setup with ISO 4217 validation
- Exchange rate management (manual entry and API integration)
- Dual-amount entry for foreign currency transactions
- IAS 21 compliant currency revaluation with automatic prior period reversal
- Unrealized vs realized gain/loss tracking and reporting
- Foreign currency sub-ledger maintaining parallel balances

Cash Management & Treasury (NEW - Critical Addition)

- Bank account management with multi-currency support
- Automatic bank reconciliation with smart matching rules
- Manual reconciliation interface for exceptions
- Bank statement import (CSV/Excel formats)
- Cash position monitoring and daily aggregation
- Cash flow forecasting (30-90 day projections)
- Imprest account management with replenishment workflows
- Cheque register and tracking

Accounts Payable

- Supplier master with credit terms and tax details
- Invoice processing with 2-way and 3-way matching to POs
- Payment processing with batch selection and authorization
- Supplier aging and payment tracking
- Early payment discount calculations
- Withholding tax handling

Accounts Receivable

- Customer master with credit limits and terms
- Invoice generation with tax calculation
- Payment receipt and allocation
- Customer aging (0-30, 31-60, 61-90, 90+ days)
- Late fee calculations
- Bad debt provisioning (basic percentage method)
- Duplicate invoice detection

Basic Tax Management (NEW - Essential Addition)

- VAT/GST rate configuration and tracking
- Automatic tax calculation on transactions
- Withholding tax setup and calculation
- Tax reports (Input VAT, Output VAT, Withholding Tax summary)
- Tax account reconciliation

Budget Management

- Budget creation by department and account
- Budget allocation (Revenue, CapEx, OpEx)
- Real-time budget consumption tracking
- Variance analysis and reporting
- Budget threshold alerts

Inventory Management (Basic)

- Item master with SKU management

- Warehouse and location tracking
- Stock movements (receipts, issues, transfers, adjustments)
- Weighted Average Cost (WAAC) valuation method
- Physical stock take with variance resolution
- Inventory valuation reports and GL integration

Fixed Assets (Simplified for MVP)

- Asset master with category hierarchy
- Asset acquisition from purchase orders
- Single-book depreciation (Corporate book only for MVP)
- Depreciation methods: Straight-Line, Reducing Balance, Sum of Years Digits
- Asset transfers between departments/locations
- Asset disposal with gain/loss calculation
- Depreciation run and posting to GL
- Asset register and roll-forward reports

Period Close & Financial Controls (NEW - Critical for Closing Books)

- Fiscal year and period definition
- Period open/close controls with validation
- Pre-close checklist (trial balance validation, bank recs complete, etc.)
- Period lock preventing backdated transactions
- Closing entry templates for accruals and deferrals
- Year-end close with retained earnings transfer

OUT OF SCOPE - Deferred to v2.0

Fixed Assets - Advanced Features

- Asset componentization (IAS 16) - Complex, defer to v2.0
- Construction Work in Progress (CWIP) and automated capitalization
- Multi-book accounting (Tax book, Statutory book)
- Asset revaluation and impairment testing (IAS 36)
- IFRS 16 lease accounting (ROU assets and liabilities)
- Physical asset tracking with barcode/RFID integration

Inventory - Advanced Features

- FIFO costing method
- Lot and serial number tracking
- Batch expiry management
- Advanced warehouse management (bin locations, pick/pack/ship)

Accounts Payable - Advanced Features

- GRNI (Goods Received Not Invoiced) accrual automation
- Purchase Price Variance (PPV) analysis
- Encumbrance accounting for purchase commitments

Currency - Advanced Features

- Fully automated revaluation scheduling (keep manual trigger for MVP)

Tax Management - Advanced Features

- Deferred tax asset/liability calculation

- Transfer pricing calculations

Budget - Advanced Features

- Overhead allocation (step-down, reciprocal methods)
- Construction budget management with phase tracking
- Activity-Based Costing (ABC)

Banking - Advanced Features

- Direct bank API integration (real-time balance, electronic transfers)
- Payment gateway integration for online collections

DETAILED DEVELOPMENT TIMELINE

PHASE 1: Foundation & Core Accounting (Weeks 1-9, 45 days)

Week 1-2: Chart of Accounts & Multi-Currency Setup (10 days)

Day 1-4: Segmented Account Structure (4 days)

- Segment configuration entity model (up to 20 segments)
- Lookup table management for segment values
- Reporting dimension flag implementation (IsReportingDimension)
- Account creation API with segment validation
- Unit tests for segment structure validation

Day 5-7: Multi-Currency Foundation (3 days)

- Currency master setup with ISO 4217 validation
- Exchange rate management API (manual and bulk upload)
- Multi-currency account linkage with protection rules
- Currency link removal validation (transaction history check)

Day 8-10: IFRS/Base/Local Classification (3 days)

- Classification framework entity model
- Account classification assignment
- Classification-based filtering for reports
- API endpoints for classification management

Week 3-5: General Ledger & Journal Entries (15 days)

Day 1-6: Journal Entry Processing (6 days)

- Journal entry entity model with header and lines
- Balanced entry validation (debits = credits)
- Multi-currency transaction processing with dual amounts
- Foreign currency sub-ledger parallel tracking
- Integration with external approval workflow API
- Posting to GL and balance updates

Day 7-12: GL Balances & Reporting (6 days)

- Account balance tracking (by period, by currency)
- Trial Balance generation with segment grouping
- Income Statement generation with comparative periods
- Balance Sheet generation with classification grouping
- Drill-down from reports to transaction detail
- Report filtering by reporting dimensions only

Day 13-15: Currency Revaluation (IAS 21) (3 days)

- Revaluation configuration (gain/loss accounts)
- Prior period revaluation detection and automatic reversal
- Current period revaluation calculation (IAS 21 formula)
- Automatic journal entry generation for unrealized gains/losses
- Revaluation history tracking and reporting

Week 6-7: Period Close & Financial Controls (10 days)

Day 1-4: Fiscal Calendar & Period Management (4 days)

- Fiscal year and period definition
- Period open/close status management
- Period lock validation on transaction posting
- API endpoints for period administration

Day 5-8: Close Checklist & Validation (4 days)

- Pre-close checklist configuration
- Trial balance validation (debits = credits)
- Bank reconciliation completion check
- Outstanding items report
- Period close API with validation enforcement

Day 9-10: Closing Entries & Year-End (2 days)

- Closing entry templates (accruals, deferrals, reclassifications)
- Year-end retained earnings transfer
- Audit trail for all close activities

Week 8-9: Testing & Integration (10 days)

- Unit tests for all Phase 1 components
- Integration testing (COA → GL → Reports → Close)
- Multi-currency transaction end-to-end testing
- Revaluation workflow testing with reversals
- Period close simulation with full month cycle
- API documentation (Swagger/OpenAPI)
- Bug fixes and refinements

PHASE 2: Cash Management & Payables (Weeks 10-14, 25 days)

Week 10-11: Cash Management & Bank Reconciliation (10 days)

Day 1-5: Bank Account Management (5 days)

- Bank account master with multi-currency support
- Bank statement import (CSV/Excel parsers) - *defer*
- Cash position dashboard with real-time aggregation - *defer*
- Daily cash balance tracking by account
- Multi-currency cash balance reporting

Day 6-10: Bank Reconciliation (5 days)

- Automatic matching rules engine (amount, date, reference) - *defer*
- Manual reconciliation interface for exceptions
- Outstanding items tracking (uncleared cheques, deposits in transit)
- Bank reconciliation report with audit trail
- Period-end bank rec completion validation

Week 12: Treasury Operations (5 days)

Day 1-3: Cash Flow Forecasting (3 days)

- Cash flow projection model (30/60/90 day horizons) - *defer*
- Expected cash inflows from AR aging
- Expected cash outflows from AP aging
- Cash forecast report with liquidity gap analysis- *defer*

Day 4-5: Imprest & Cheque Management (2 days)

- Imprest account setup with limits and custodians
- Imprest replenishment workflow (retirement and top-up)
- Cheque register and tracking
- Cheque printing format templates

Week 13-14: Accounts Payable (10 days)

Day 1-4: Supplier Management & Invoicing (4 days)

- Supplier master with payment terms and bank details
- Supplier invoice capture and validation
- 2-way matching (invoice vs. purchase order)
- 3-way matching (invoice vs. PO vs. goods receipt)
- Invoice approval routing to external workflow API

Day 5-8: Payment Processing (4 days)

- Payment batch creation with due date filtering
- Early payment discount calculation and selection
- Payment authorization workflow
- Payment posting to cash accounts
- Withholding tax calculation
- Payment reconciliation with bank statements

Day 9-10: AP Reporting & Analytics (2 days)

- Supplier aging report (0-30, 31-60, 61-90, 90+ days)
- Cash requirement forecast from AP aging
- Payment history and supplier statement
- Withholding tax summary reports

PHASE 3: Receivables & Tax (Weeks 15-17, 15 days)

Week 15-16: Accounts Receivable (10 days)

Day 1-4: Customer Management & Invoicing (4 days)

- Customer master with credit limits and payment terms
- Invoice generation API with line items
- Automatic tax calculation (VAT/GST)
- Invoice numbering with prefix/suffix configuration
- Duplicate invoice detection

Day 5-8: Payment Receipt & Allocation (4 days)

- Payment receipt capture (bank, cash, mobile money, online)
- Automatic invoice matching by customer and amount- *defer*
- Manual payment allocation for partial or multi-invoice payments
- Overpayment and credit note handling
- Receipt printing templates

Day 9-10: AR Analytics & Collections (2 days)

- Customer aging report (0-30, 31-60, 61-90, 90+ days)
- Late fee calculation and posting- *defer*
- Bad debt provisioning (percentage of aged balances) - *defer*
- Customer statement generation
- Collections dashboard with overdue highlights

Week 17: Basic Tax Management (5 days)

Day 1-3: Tax Configuration & Calculation (3 days)

- Tax rate setup (VAT, GST, withholding tax)
- Tax applicability rules (by product, customer, supplier)
- Automatic tax calculation on AR and AP transactions
- Tax account posting and tracking

Day 4-5: Tax Reporting (2 days)

- Input VAT report (VAT paid on purchases)
- Output VAT report (VAT collected on sales)
- VAT reconciliation and net payable/receivable
- Withholding tax summary by supplier and type
- Withholding tax certificate data export

PHASE 4: Inventory, Budget & Fixed Assets (Weeks 18-22, 25 days)

Week 18-19: Inventory Management (10 days)

Day 1-4: Item Master & Stock Movements (4 days)

- Item master with SKU, description, category, UOM
- Warehouse and location setup
- Stock receipt from purchase orders
- Stock issues to production or sales
- Stock transfers between warehouses
- Stock adjustments with reason codes

Day 5-8: WAC Costing & Valuation (4 days)

- Weighted Average Cost calculation engine
- Cost layer tracking for receipts
- Cost allocation on issues
- GL integration (inventory account, COGS posting)
- Inventory valuation reports

Day 9-10: Stock Take & Reporting (2 days)

- Physical stock count sheet generation
- Count entry and variance calculation
- Stock adjustment posting from count variances
- Stock movement report
- Stock level alerts (reorder point, overstock)

Week 20: Budget Management (5 days)

Day 1-3: Budget Setup & Allocation (3 days)

- Fiscal year and budget period definition
- Budget allocation by department and account
- Budget categories (Revenue, CapEx, OpEx)
- Budget entry API with bulk upload support

- Budget approval workflow integration

Day 4-5: Budget Tracking & Reporting (2 days)

- Real-time budget consumption calculation- *defer*
- Budget threshold alerts (80%, 90%, 100%)- *defer*
- Budget vs. actual variance analysis
- Budget utilization dashboard

Week 21-22: Fixed Assets (Simplified) (10 days)

Day 1-4: Asset Master & Acquisition (4 days)

- Asset master with category, location, custodian
- Asset category setup with default depreciation parameters
- Asset acquisition from purchase orders - *defer*
- Capitalizable cost allocation
- GL posting for asset capitalization

Day 5-8: Depreciation Processing (4 days)

- Depreciation methods: Straight-Line, Reducing Balance, Sum of Years Digits
- Monthly depreciation calculation engine
- Depreciation run execution and preview
- Automatic journal entry generation for depreciation
- Depreciation schedule reports

Day 9-10: Asset Transfers & Disposal (2 days)

- Asset transfer between departments/locations
- Asset disposal with sale proceeds
- Gain/loss on disposal calculation
- Disposal journal entry generation
- Asset register and roll-forward reports
- Net book value tracking by asset

PHASE 5: Integration, Testing & Documentation (Weeks 23-25, 15 days)

Week 23: End-to-End Integration Testing (5 days)

- Complete business cycle testing (Purchase → AP → Payment → Bank Rec)
- Sales cycle testing (Invoice → AR → Receipt → Bank Rec)
- Inventory cycle testing (Receipt → Stock → Issue → COGS)
- Fixed asset lifecycle testing (Acquisition → Depreciation → Disposal)
- Month-end close simulation with all modules
- Multi-currency transaction flows including revaluation
- Report generation from all major modules

Week 24: Performance Testing & Optimization (5 days)

- Load testing with realistic data volumes
- API response time optimization
- Database query optimization and indexing
- Report generation performance tuning
- Concurrent user simulation

Week 25: Documentation & Deployment Preparation (5 days)

- Complete API documentation (Swagger/OpenAPI with examples)
- Integration guide for external developers

- Admin configuration guide
- Database schema documentation
- Deployment scripts and CI/CD pipeline setup
- Security audit and vulnerability assessment
- MVP go-live checklist and readiness assessment

EFFORT SUMMARY

Module/Component	Days	Notes
Chart of Accounts & Segmented Structure	10	Includes reporting dimensions
General Ledger & Journal Entries	15	Multi-currency + revaluation
Period Close & Financial Controls	10	Critical for closing books
Phase 1 Testing & Integration	10	End-to-end GL cycle
PHASE 1 SUBTOTAL	45	Weeks 1-9
Cash Management & Bank Reconciliation	10	Essential for treasury
Treasury Operations	5	Forecasting + imprest
Accounts Payable	10	Full AP cycle
PHASE 2 SUBTOTAL	25	Weeks 10-14
Accounts Receivable	10	Full AR cycle
Basic Tax Management	5	VAT + withholding tax
PHASE 3 SUBTOTAL	15	Weeks 15-17
Inventory Management	10	WAC only for MVP
Budget Management	5	Core budgeting
Fixed Assets (Simplified)	10	Single-book, core methods
PHASE 4 SUBTOTAL	25	Weeks 18-22
Integration Testing	5	End-to-end scenarios
Performance Testing	5	Load testing + optimization
Documentation & Deployment	5	API docs + deployment prep
PHASE 5 SUBTOTAL	15	Weeks 23-25
TOTAL MVP EFFORT	125	25 weeks / 4.5 months

Working Assumptions:

- Approval workflow API already available and tested
- RHEMA ERP infrastructure (auth, multi-tenancy, repository pattern) fully functional
- Database and deployment infrastructure already provisioned

RISK ASSESSMENT & MITIGATION

Risk Factor	Impact	Mitigation
Currency Revaluation Complexity	Auto-reversal logic may take longer than estimated	3-day buffer in Phase 1. Simplify if needed for MVP
Bank Reconciliation Edge Cases	Complex matching scenarios may delay testing	Start with simple rules, enhance iteratively
Fixed Assets Scope Creep	Stakeholders may demand multi-book or CWIP	Strict MVP definition. Defer to v2.0 with documentation
Performance Issues at Scale	Reports slow with 10K+ transactions	Dedicated performance testing week. DB optimization
Integration Testing Delays	Cross-module issues discovered late	Continuous integration testing after each phase
Approval API Changes	External API contract breaks or changes	Early integration testing. Contract versioning
Customer Validation Failure	MVP missing must-have features for sales	Validate scope at Week 12 milestone with stakeholders

Overall Risk Level: Medium - Achievable with disciplined execution and clear scope boundaries.

Risk Reduction Factors:

- Realistic effort estimates based on actual workflow complexity
- Clearly defined MVP scope with documented v2.0 deferrals
- Phase-based delivery with testing after each phase
- Simplified Fixed Assets (single-book) reduces complexity significantly
- Approval workflow handled externally (not in critical path)
- Dedicated time for performance testing and optimization

POST-MVP ROADMAP (v2.0 - v3.0)

Quarter 1 Post-Launch (Months 5-7) - v2.0

- **Fixed Assets Enhancements:**
 - Multi-book depreciation (Tax book, Statutory book) (~12 days)
 - Deferred tax calculation on timing differences (~5 days)
 - Asset componentization framework (IAS 16) (~8 days)
 - Asset revaluation and revaluation reserve (~6 days)

- **Advanced Inventory Features:**
- FIFO costing method (~8 days)
- Lot and serial number tracking (~6 days)
- **Accounts Payable Automation:**
- GRNI accrual automation (~4 days)
- Purchase Price Variance (PPV) (~3 days)

Quarter 2 Post-Launch (Months 8-10) - v2.5

- **Fixed Assets Advanced:**
- CWIP and automated capitalization (~10 days)
- Asset impairment testing (IAS 36) (~8 days)
- Physical asset tracking with barcode/RFID (~10 days)
- **Advanced Budget Features:**
- Overhead allocation (step-down method) (~8 days)
- Construction budget with phase tracking (~6 days)

Quarter 3 Post-Launch (Months 11-13) - v3.0

- **IFRS 16 Lease Accounting:**
- Right-of-Use (ROU) asset recognition (~12 days)
- Lease liability calculation and amortization (~8 days)
- Short-term and low-value lease exemptions (~4 days)
- **Tax Automation:**
- GRA eTax filing integration (~12 days)
- Deferred tax asset/liability full automation (~8 days)
- Transfer pricing calculations (~6 days)
- **Banking Integration:**
- Direct bank API integration for Ghana banks (~10 days)
- Mobile money integration (MTN, Vodafone, AirtelTigo) (~12 days)
- Payment gateway for online collections (~8 days)

MVP SUCCESS CRITERIA

Close books monthly for internal operations with full audit trail

Market-ready API with complete documentation

Complete financial cycle (GL, AP, AR, Cash)

Multi-currency with IAS 21 compliant revaluation

Cash management and bank reconciliation

Basic inventory (WAAC) and budget management

Core fixed assets with standard depreciation methods

Period close procedures with validation and controls

KEY IMPROVEMENTS IN THIS REVISION

Added 15 days for Cash Management/Treasury (critical omission)

Added 10 days for Period Close procedures (essential for month-end)

Added 5 days for Basic Tax Management (VAT + withholding)

Properly scoped Fixed Assets (removed multi-book, CWIP, componentization)

Enhanced Currency Revaluation with auto-reversal (IAS 21 compliant) - *defer*

Realistic effort estimates based on comprehensive workflow analysis

Clear MVP vs. v2.0 boundaries to prevent scope creep

END OF REVISED REALISTIC 4.5-MONTH MVP PLAN

Total Effort: 125 Days | 25 Weeks | 4.5 Months